
NEWS TOPIC CLASSIFICATION USING CLASSICAL MACHINE LEARNING: SPARSE FEATURES VS CONTEXTUAL EMBEDDINGS

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Statistical Machine Learning Project

ABSTRACT

This project explores news topic classification using classical machine learning techniques on different textual feature spaces. Sparse representations such as Bag-of-Words and TF-IDF were compared with contextual embeddings generated using DistilBERT. Dimensionality reduction techniques including Truncated SVD and PCA were also analyzed to study their impact on efficiency and classification performance.

Keywords NLP, News Classification, TF-IDF, DistilBERT, PCA, SVD, t-SNE

1 Introduction

Text classification is a fundamental task in Natural Language Processing (NLP), where the goal is to assign predefined categories to textual data. Traditionally, machine learning models relied on sparse representations such as Bag-of-Words (BoW) and TF-IDF. However, recent advancements have introduced contextual embeddings like BERT that capture semantic meaning.

In this project, we compare classical machine learning models using two types of feature representations:

- Sparse representations (BoW and TF-IDF)
- Contextual embeddings (DistilBERT)

The objective is to evaluate how feature representation impacts classification performance.

2 Dataset Description and EDA

We used the AG News dataset, which consists of news articles categorized into four classes:

- World
- Sports
- Business
- Sci/Tech

From the dataset, we selected:

- 8,000 training samples
- 2,000 test samples

EDA Performed:

- Class distribution plot
- Text length distribution
- Sample text inspection
- t-SNE visualization

Observations:

- The dataset is fairly balanced across classes
- Most articles contain short to medium-length text
- BERT embeddings produced clearer semantic clusters than sparse TF-IDF features

3 Methodology

3.1 Feature Representations

3.1.1 (a) Bag-of-Words (BoW)

- Converts text into word frequency vectors
- Uses unigrams
- Stopwords removed

3.1.2 (b) TF-IDF

- Assigns importance based on word frequency and rarity
- Uses unigrams + bigrams
- Stopwords removed

3.1.3 (c) Contextual Embeddings (DistilBERT)

- Used DistilBERT as a feature extractor
- Extracted CLS token embeddings
- Embeddings stored as NumPy arrays
- No fine-tuning performed

3.2 Dimensionality Reduction

- Truncated SVD applied on TF-IDF features
- PCA applied on BERT embeddings

Purpose of dimensionality reduction:

- Reduce dimensionality
- Improve efficiency
- Remove noise

3.3 Models Used

The following classical ML models were trained:

- Logistic Regression
- Support Vector Machine (Linear and RBF)
- K-Nearest Neighbors (KNN)
- KMeans (unsupervised baseline)

4 Experiments

Each model was trained on the following feature sets:

1. Bag-of-Words features
2. TF-IDF features
3. TF-IDF + SVD
4. BERT embeddings
5. BERT embeddings + PCA

Evaluation metrics used:

- Accuracy
- Macro F1 Score
- Weighted F1 Score

Experiments performed:

- Accuracy comparison
- F1 score comparison
- SVD explained variance analysis
- PCA explained variance analysis
- Confusion matrix generation
- t-SNE visualization
- Real-time text inference

5 Results

Key Observations:

- BERT + SVM (RBF) achieved the highest Accuracy, highest Macro F1 Score, and highest Weighted F1 Score among all models.
- TF-IDF consistently outperformed Bag-of-Words in Accuracy, Macro F1 Score, and Weighted F1 Score due to better feature weighting.
- DistilBERT embeddings produced stronger semantic representations, resulting in improved classification performance across all evaluation metrics.
- KNN performance improved significantly with TF-IDF and DistilBERT features compared to sparse Bag-of-Words representations.
- SVD and PCA reduced feature dimensionality while preserving most of the classification performance.
- KMeans produced the lowest Accuracy, Macro F1 Score, and Weighted F1 Score because it is an unsupervised clustering algorithm.

6 Discussion

The results clearly show that:

- Sparse representations (BoW, TF-IDF) are effective but limited
- Contextual embeddings provide better semantic understanding
- Dimensionality reduction helps improve performance and efficiency

Why DistilBERT Performs Better:

- Captures word meaning in context
- Understands sentence structure
- Generates semantically meaningful embeddings

Limitations:

- BERT embeddings require more computation
- No fine-tuning was performed
- t-SNE visualization is computationally expensive

7 Conclusion and Learnings

This project demonstrates that:

- Feature representation plays a crucial role in NLP tasks
- Classical ML models can perform well with strong features
- Contextual embeddings outperform traditional methods

Key Learnings:

- Importance of preprocessing
- Role of dimensionality reduction
- Comparison of feature engineering techniques
- Importance of contextual embeddings in NLP
- Visualization of high-dimensional data using t-SNE

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